

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO4: Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5)

Assessment Tool Weightage:20%

QUESTIONS: 5

Total Marks:50

Annexure A

Industry Problem Solving Task

Code of Conduct:

I M. Phanith Chandu (192425238) Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M. Phanith Chandu

Q. No	Understanding of Industry Problem (20%)	Application of Engineering Concepts (25%)	Solution Design (25%)	Use of Tools (15%)	Analysis (10%)	Professionalism Total Score (5%) (100%)	
1							
2							
3							
4							
5							
Total Marks (100)				Total Marks (50)			

Annexure A

Industry Problem Solving Task

Analyze the given problem by comparing available techniques against the specified constraints and requirements. Evaluate and justify your decision using logical reasoning, clearly indicating how the chosen approach satisfies the conditions.

1. Smart Surveillance System Selection

A city surveillance system must operate under the following conditions:

- If latency > 100 ms $\rightarrow$ system fails
- If accuracy $< 85\%$ $\rightarrow$ unacceptable
- Hardware supports only moderate computation

Models available:

Viola-Jones: 78%, low latency

YOLO: 92%, moderate latency

Deep Learning: 95%, high latency

Compare the models and evaluate which satisfies all constraints. Justify your decision using logical reasoning based on given conditions.

2. Autonomous Vehicle Detection Logic

A vehicle detection system must:

- Prioritize safety (accuracy $> 90\%$)
- Maintain response time < 50 ms

Options:

Method A: 88% accuracy, 30 ms

Method B: 93% accuracy, 70 ms

Method C: 91% accuracy, 45 ms

Compare the methods and evaluate which should be selected. Justify your choice using constraint-based reasoning.

3. Motion Analysis Decision Problem

A traffic system requires:

- Accurate motion direction detection
- Robustness in dense traffic

Results:

Optical Flow: accurate but noisy in dense scenes

Motion Estimation: stable but less detailed

Compare both approaches and evaluate which is suitable under high-density conditions. Justify using scenario-based logic.

4. Background Modeling Evaluation

A GMM-based system produces:

- Static scenes: 90% accuracy

- Dynamic scenes: 65% accuracy Requirement:

System must maintain $\geq 80\%$ accuracy across all scenarios

Compare performance and evaluate whether GMM alone meets requirements. Justify logically and suggest a decision.

5. Defect Detection System Decision

Requirements:

- Detect subtle defects ($>95\%$ accuracy)
- Adapt to product variations

Options:

Traditional methods: fast, less adaptive

Deep learning: accurate, requires training

Compare both approaches and evaluate which meets requirements. Justify using performance vs adaptability logic.

1. Smart Surveillance

System Selection

Given

constraints:

- Latency must not exceed 100 ms; otherwise, the system fails.
- Accuracy must be at least 85%; otherwise, the result is unacceptable.
- The available hardware supports only moderate computation.

Model comparison:

Model	Accuracy	Latency	Computation
Viola-Jones	78%	Low	Low
YOLO	92%	Moderate	Moderate
Deep Learning	95%	High	High

Understanding of Industry Problem

- The surveillance system must satisfy accuracy, latency, and hardware-computation constraints simultaneously.
- An accuracy below 85% is unacceptable.
- Latency above 100 ms causes system failure.
- Because the hardware supports only moderate computation, a high-computation method is not ideal.

Analysis

- Viola-Jones has low latency and low computational demand, but its 78% accuracy is below the required 85%, so it fails the accuracy constraint.
- Deep Learning has the highest accuracy at 95%, but it requires high computation and has high latency, making it unsuitable for the given hardware and latency constraints.
- YOLO provides 92% accuracy, which is above the 85% minimum requirement.
- YOLO has moderate latency, which is more compatible with the 100 ms limit than the high-latency Deep Learning option.
- YOLO also requires moderate computation, matching the stated hardware capability.

Application of Engineering Concepts

- This is a constraint-based model-selection problem: accuracy, latency, and computational cost must all be considered.
- A surveillance system needs both accurate detection and sufficiently fast response.
- Using a model that exceeds the hardware capability can reduce real-world performance even if its accuracy is higher.
- YOLO provides a practical balance between detection accuracy and computational requirements.

Solution Design

- Select YOLO as the primary detection model.
- Deploy YOLO on the moderate-computation hardware.
- Validate actual latency during deployment to ensure it remains below 100 ms.
- Monitor accuracy using representative surveillance data and adjust the model or input settings if required.

Use of Tools

- Use a computer-vision framework such as OpenCV or a YOLO implementation for model deployment.
- Use a test video dataset containing normal surveillance conditions to evaluate detection accuracy.
- Measure inference latency using the target hardware before deployment.
- Use performance logs to compare accuracy and response time.

Conclusion

YOLO is the best choice because it satisfies the stated accuracy requirement, is compatible with moderate computation, and has moderate latency. Viola-Jones fails because its accuracy is only 78%, while Deep Learning is unsuitable because its computation and latency are high.

2. Autonomous Vehicle Detection Logic

Given constraints:

- Safety is the priority, so accuracy must be greater than 90%.
- Response time must be less than 50 ms.

Method comparison:

Method	Accuracy	Response Time
Method A	88%	30 ms
Method B	93%	70 ms
Method C	91%	45 ms

Understanding of Industry Problem

- The autonomous vehicle must make safe detection decisions.
- Accuracy must be greater than 90%, not merely equal to 90%.
- Response time must remain below 50 ms.
- Both conditions must be satisfied because failure in either accuracy or response time can make the method unsuitable.

Analysis

- Method A responds in 30 ms, but its accuracy is only 88%, so it fails the safety accuracy requirement.
- Method B has 93% accuracy, satisfying the accuracy requirement, but its response time is 70 ms, which fails the 50 ms limit.
- Method C has 91% accuracy and a 45 ms response time.
- Method C satisfies both constraints simultaneously.
- Therefore, Method C provides the best constraint-based solution.

Application of Engineering Concepts

- The decision uses hard constraints rather than choosing the method with the highest accuracy alone.
- In autonomous vehicles, both detection accuracy and response time are critical because delayed information can affect driving decisions.
- A method with excellent accuracy but excessive latency may still be unsafe.
- A method with fast response but insufficient accuracy may also create unsafe detections.

Solution Design

- Select Method C for the vehicle detection system.
- Target an accuracy above 90% and maintain response time below 50 ms during deployment.
- Test Method C under different traffic and environmental conditions.
- Monitor both accuracy and latency continuously during system validation.

Use of Tools

- Use a computer-vision testing framework to evaluate detection accuracy.
- Use real-time video or recorded driving datasets for validation.
- Measure inference time on the actual vehicle hardware.
- Record false detections, missed detections, and response times for performance evaluation.

Conclusion

Method C should be selected. It achieves 91% accuracy, which is above the required 90%, and

responds in 45 ms, which is below the 50 ms limit. Method A fails accuracy, while Method B fails response time.

3. Motion Analysis Decision Problem

Industry requirements:

- Accurate motion-direction detection.
- Robustness in dense traffic.

Given results:

Approach	Strength	Limitation
Optical Flow	Accurate motion direction	Noisy in dense scenes
Motion Estimation	Stable	Less detailed

Understanding of Industry Problem

- The traffic system must identify motion direction accurately.
- The system must remain robust when many vehicles are present at the same time.
- Dense traffic can create overlapping motion patterns and increase noise.
- Therefore, both accuracy of motion information and stability in crowded scenes are important.

Analysis

- Optical Flow gives accurate motion information and can capture detailed movement.
- However, its vectors become noisy in dense traffic, reducing robustness.
- Motion Estimation provides more stable results, which is useful in crowded traffic.
- Its limitation is that it provides less detailed motion information.
- Because the problem specifically emphasizes robustness in dense traffic, stability becomes an important factor.

Application of Engineering Concepts

- Dense traffic contains multiple moving objects whose motion vectors may overlap.
- A stable motion-estimation approach can reduce the effect of noisy motion information.
- Optical Flow may still be useful when detailed motion direction is more important and the scene is less crowded.
- A hybrid approach can also be considered by using detailed optical-flow information with filtering or object-level tracking.

Solution Design

- For high-density traffic, select Motion Estimation as the safer primary approach because it is

more stable.

- Apply filtering and object tracking to maintain reliable motion information.
- Use Optical Flow as an additional detailed-motion module when needed.
- Validate the selected approach using traffic videos with different vehicle densities.

Use of Tools

- Use OpenCV or a similar computer-vision framework to implement motion analysis.
- Use traffic video datasets containing low-, medium-, and high-density traffic.
- Measure direction-detection accuracy and stability of motion vectors.
- Visualize motion vectors to identify noisy or unstable regions.

Conclusion

For high-density traffic, Motion Estimation is more suitable because stability and robustness are important in crowded scenes. Optical Flow is more detailed and accurate in suitable conditions, but its noise in dense traffic makes it less reliable as the sole method. A hybrid design can provide additional detail when required.

4. Background Modeling Evaluation

Given results and requirement:

Scenario	GMM Accuracy	Requirement
Static scenes	90%	$\geq 80\%$
Dynamic scenes	65%	$\geq 80\%$

Understanding of Industry Problem

- The system must maintain at least 80% accuracy in every scenario.
- GMM achieves 90% in static scenes.
- GMM achieves only 65% in dynamic scenes.
- Therefore, performance must be evaluated across both static and dynamic environments.

Analysis

- In static scenes, GMM exceeds the requirement by 10 percentage points.
- In dynamic scenes, GMM is 15 points below the required 80%.
- The large performance difference shows that GMM is sensitive to background movement.
- Since the requirement applies across all scenarios, passing only the static-scene condition is not sufficient.

- Therefore, GMM alone does not meet the overall requirement.

Application of Engineering Concepts

- Background modeling must adapt to changes such as moving trees, vehicles, shadows, and other dynamic background elements.
- Adaptive background updating can help the model learn changing background patterns.
- Threshold tuning and morphological filtering can reduce false detections.
- A complementary object-detection method can improve performance when background subtraction becomes unreliable.

Solution Design

- Do not deploy the existing GMM configuration as the only detection method.
- Improve GMM using adaptive updating and suitable parameter tuning.
- Test the improved system specifically on dynamic scenes because this is where the requirement is currently violated.
- Use a hybrid detection pipeline if GMM still cannot achieve at least 80% accuracy.

Use of Tools

- Use OpenCV or another vision library to implement and tune GMM background subtraction.
- Evaluate the model using both static and dynamic traffic or surveillance videos.
- Measure accuracy separately for each scenario.
- Use validation results to select thresholds and update parameters.

Conclusion

GMM alone does not satisfy the requirement because its dynamic-scene accuracy is only 65%, below the required 80%. It performs well in static scenes, but the system needs adaptive modeling, parameter tuning, preprocessing, or a complementary detection method to meet the requirement across all scenarios.

5. Defect Detection System Decision

Requirements:

- Subtle defects must be detected with accuracy greater than 95%.
- The system must adapt to product variations.

Option comparison:

Approach	Performance	Adaptability
Traditional methods	Fast	Less adaptive
Deep Learning	Accurate	Requires training

Understanding of Industry Problem

- The system must detect subtle defects with very high accuracy.
- The system must also handle changes in product appearance or variation.
- Fast processing alone is not enough because the accuracy requirement is strict.
- Adaptability is also important for industrial inspection where product variations may occur.

Analysis

- Traditional methods are fast, but the problem states that they are less adaptive.
- This lack of adaptability can reduce performance when product appearance changes.
- Deep Learning is described as accurate, making it more suitable for subtle-defect detection.
- Deep Learning requires training, so sufficient representative training data is necessary.
- Because the requirement emphasizes accuracy above 95% and adaptation to product variations, Deep Learning is the stronger option.

Application of Engineering Concepts

- Deep Learning can learn visual patterns associated with defects from labeled training examples.
- Training data should include different product variations, defect types, sizes, positions, and appearances.
- Validation data should be kept separate from training data to evaluate generalization.
- The model should be retrained or updated when new product variations appear.

Solution Design

- Select a Deep Learning-based inspection system.
- Build a representative dataset covering normal products and subtle defects across product variations.
- Train and validate the model using separate datasets.
- Measure accuracy against the required threshold of greater than 95%.
- Deploy the model with periodic monitoring and retraining when product variations change.

Use of Tools

- Use a deep-learning framework such as TensorFlow or PyTorch for model development.
- Use image-collection and annotation tools to create a labeled defect dataset.
- Use validation metrics such as accuracy, precision, recall, and confusion matrix analysis.
- Use industrial inspection images from different product batches for testing.